DataConstants

+ USER_FILE_PATH: String = "../json/user.json"
+ SHELTER_FILE_PATH: String = "../json/shelter.json"
+ REQUEST_FILE_PATH: String = "../json/requests.json"
+ HURRICANE_FILE_PATH: String = "../json/hurricane.json"
...

DataReader

+ getUsers(): HashMap<String, User>
+ getShelters(): Hashmap<String, Shelter>
+ getHurricanes(): HashMap<String,Hurricane>
+ getRequests(): HashMap<UUID, Request>

DataWriter

+ saveUsers(HashMap<String, User> users): boolean
+ saveShelters(HashMap<String, Shelter> shelters): boolean
+ saveHurricanes(HashMap<String, Hurricane> hurricanes): boolean
+ saveRequests(HashMap<UUID, Request> requests): boolean

HurricaneUI

- CARTER_USERNAME: String
- CARTER_PASSWORD: String
- ROSA_USERNAME: String
- ROSA_PASSWORD: String
- FELIX_USERNAME: String
- FELIX_PASSWORD: String
- TRIP_FILE: String
- app: HurricaneApplication
- carterRequest: Request
- chosenShelter: Shelter

~ HurricaneUI()
+ run(): void
+ createReliefRequest(): void
+ matchRosa(): void
+ findShelterAndAssign(): void
+ exportTrip(): void
+ rolePermissions(): void
+ checkInFamily(): void
+ verify(): void
- printDirections(String title, String[] directions): void
- printShelter(Shelter shelter): void
- accomodationText(ArrayList<Accomodation> accomodations): String
- namesText(Collection<String> names): String
- summarize(Request request): String
- heading(String title): void
+ main(String[] args): void

UserList

- userList: UserList
- users: HashMap<String, User>

- UserList()
+ getInstance(): UserList
+ getUser(String name): User
+ addUser(User user): void
+ save(): boolean
+ getUserList(): HashMap<String, User>
+ usernameTaken(String username): boolean

RequestList

- requestList: RequestList
- requests: HashMap<UUID, Request>

- RequestList()
+ getInstance(): RequestList
+ getRequest(UUID id): Request
+ addRequest(Request request): void
+ save(): boolean
+ getRequestList(): HashMap<UUID, Request>

ShelterList

- shelterList: ShelterList
- shelters: HashMap<String, Shelter>

- ShelterList()
+ getInstance(): ShelterList
+ getShelter(String name): Shelter
+ addShelter(Shelter shelter): void
+ save(): boolean
+ getShelterList(): HashMap<String, Shelter>

HurricaneList

- hurricaneList: HurricaneList
- hurricanes: HashMap<String, Hurricane>

- HurricaneList()
+ getInstance(): HurricaneList
+ getHurricane(String name): Hurricane
+ addHurricane(Hurricane hurricane): void
+ save(): boolean
+ getHurricaneList(): HashMap<String, Hurricane>

Address

- street: String
- city: String
- state: String
- zip: String

+ Address(String street, String city, String state, String zip)
+ Address(String city, String state)
+ getCoordinates(): Coordinates
+ getDistanceTo(Address address): double
+ isWithin(Address address, int distance): boolean
+ toString(): String

Request

- id: UUID
- requestee: String
- severity: byte
- address: Address
- status: ResponseStatus
- responderType: ResponseType
- description: String
- MedicalNote: String

+ Request(UUID id, String requestee, byte severity, Address address, ResponseStatus status, ResponseType responderType, String description, String MedicalNote)
+ Request(String requestee, byte severity, Address address, String description)
+ editSeverity(byte severity): void
+ updateStatus(ResponseStatus status): void
+ toString(): String
+ getRequesteeUser(): User
+ exportToTxt(String fileName, User responder, Shelter shelter): boolean
+ toPrintableText(User responder, Shelter shelter): String
- appendDirections(StringBuilder text, String divider, String heading, Address from, Address to): void
- ageText(int age): String
- describe(String description): String
- isBlank(String text): boolean

<<enumeration>> ResponseStatus

IN_PROGRESS
WAITING
MORE_INFO_NEEDED
COMPLETE
ASSIGNED

<<enumeration>> ResponseType

MEDICAL
RESOURCE
ASSISTANCE
TRAVEL

<<enumeration>> HurricaneStatus

INCOMING
IN_PROGRESS
OVER

Hurricane

- id: UUID
- name: String
- category: byte
- currentAddress: Address
- predictedPath: ArrayList<Address>
- impactArea: ArrayList<Address>
- status: HurricaneStatus

+ Hurricane(UUID id, String name, byte category, Address currentAddress, ArrayList<Address> predictedPath, ArrayList<Address> impactArea, HurricaneStatus status)
+ Hurricane(String name, byte category, Address currentAddress)
+ addImpactArea(Address address): void
+ updateAddress(Address address): void
+ updatePredictedPath(ArrayList<Address> path): void
+ willReach(Address address): boolean
+ endHurricane(): void
+ toString(): String

HurricaneApplication

- NEARBY_MILES: int = 5
- AFFECTED_MILES: int = 25
- app: HurricaneApplication
- currentUser: User
- requestDistances: HashMap<UUID, Double>

- HurricaneApplication()
+ getInstance(): HurricaneApplication
+ login(String username, String password): boolean
+ logout(): void
+ getCurrentUser(): User
+ isType(UserType type): boolean
+ createAccount(String firstName, String lastName, String username, String emailAddress, String password): boolean
+ createAccount(String firstName, String lastName, String username, String emailAddress, String password, Address address): boolean
+ addInfo(int age, char sex, String phoneNumber, String description): boolean
+ addPartyMember(String name, int age, boolean isPet, String description): boolean
+ newRequest(byte severity, ResponseType type, String description, String medicalNote): Request
+ getRequest(UUID id): Request
+ getOpenRequests(): ArrayList<Request>
+ isRecommended(Request request): boolean
+ getRecommendationReason(Request request): String
+ getDistanceTo(Request request): double
+ getRequester(Request request): User
+ getParty(Request request): ArrayList<Person>
+ assignRequest(Request request): boolean
+ getAssignedVolunteer(Request request): Volunteer
+ finishRequest(Request request): boolean
+ exportTripInfo(Request request, Shelter shelter, String fileName): boolean
+ getDirections(Address from, Address to): String[]
+ getShelters(): ArrayList<Shelter>
+ getShelter(String name): Shelter
+ getShelterNeeds(Request request): ArrayList<Accomodation>
+ findShelters(Request request): ArrayList<Shelter>
+ findClosestShelter(Request request): Shelter
+ getMyShelter(): Shelter
+ checkIn(Request request): boolean
+ getHurricane(String name): Hurricane
+ isAffected(): boolean
+ saveData(): boolean
- setCurrentUser(User user): void
- matchesSpecialty(Request request): boolean
- distanceBetween(Address from, Address to): double
- sortableMiles(Double miles): double
- getPartySize(Request request): int
- isBlank(String text): boolean

Shelter

- id: UUID
- name: String
- address: Address
- totalCapacity: int
- usedCapacity: int
- expectedArrivals: int
- resources: ArrayList<Resource>
- accomodations: ArrayList<Accomodation>
- registeredPeople: HashMap<String, Person>
- registeredPets: HashMap<String, Person>

+ Shelter(UUID id, String name, Address address, int totalCapacity, int usedCapacity, int expectedArrivals, ArrayList<Resource> resources, ArrayList<Accomodation> accomodations, HashMap<String, Person> registeredPeople, HashMap<String, Person> registeredPets)
+ addResource(Resource resource): void
+ updateResource(Resource resource, int quantity): void
+ recieveResource(Resource resource): void
+ availableSpace(): int
+ isOutOfSupplies(): boolean
+ notify(int partySize): void
+ registerArrival(int partySize, ArrayList<Person> party): void
+ unregisterArrival(int partySize, ArrayList<Person> party): void
+ toString(): String

<<enumeration>> Accomodation

PET_FRIENDLY
MEDICAL_STAFF

Resource

- type: ResourceType
- description: String
- quantity: int
- unit: Unit

+ Resource(ResourceType type, String description, int quantity, Unit unit)
+ updateQuantity(int newQuantity): void
+ toString(): String

<<enumeration>> ResourceType

WATER
FOOD
MEDICAL
CLOTHES
BABY_NEEDS
OTHER

<<enumeration>> Unit

GALLONS
POUNDS
CANS
QTY
MEALS

Credential

- type: String
- issuer: String
- expirationDate: Date
- verified: boolean

+ Credential(String type, String issuer, Date expirationDate, boolean verified)

Alert

- affectedAddresss: ArrayList<Address>
- description: String
- issuedBy: String
- timestamp: Date

+ Alert(ArrayList<Address> affectedAddress, String description, String issuedBy, Date timestamp)
+ isRelevantTo(User user): boolean

User

User

- id: UUID
firstName: String
lastName: String
username: String
emailAddress: String
password: String
address: Address
receivedAlerts: ArrayList<Alert>
type: UserType

+ User(UUID id, String firstName, String lastName, String username, String emailAddress, String password, Address address, ArrayList<Alert> receivedAlerts)
+ User(String firstName, String lastName, String username, String emailAddress, String password, Address address)
+ User()
+ isMatch(String username, String password): boolean
+ receiveAlert(Alert alert): void
+ donate(Resource donation, Shelter shelter): void

UserManager

- scanner: Scanner

+ editUser(User user): User
+ changeUsername(User user, String username): User
+ changePassword(User user, String password): User
+ changeEmailAddress(User user, String emailAddress): User
+ changeFirstName(User user, String firstName): User
+ changeLastName(User user, String lastName): User
+ changeAddress(User user, Address address): User

<<enumeration>> UserType

VICTIM
VOLUNTEER
ADMIN
DISPATCHER
SHELTER_WORKER

Volunteer

- credentials: ArrayList<Credential>
- fieldsOfExpertise: ArrayList<ResponseType>
- identityVerified: boolean
- availible: boolean
- assignedRequest: Request

+ Volunteer(String firstName, String lastName, String username, String emailAddress, String password, Address address)
+ Volunteer(User user)
+ Volunteer(UUID id, String firstName, String lastName, String username, String emailAddress, String password, Address address, ArrayList<Alert> receivedAlerts, ArrayList<Credential> credentials, ArrayList<ResponseType> fieldsOfExpertise, boolean identityVerified, boolean availible, Request assignedRequest)
+ setAvailible(boolean available): void
+ verifyIdentity(): void
+ submitCredential(Credential credential): void
+ offerToHelp(): void
+ updateRequest(Request request, ResponseStatus status): void
+ finishRequest(Request request): void
+ toString(): String

Victim

- age: int
- sex: char
- description: String
- partySize: int
- createdRequests: ArrayList<Request>
- party: ArrayList<Person>
- phoneNumber: String

+ Victim(String firstName, String lastName, String username, String emailAddress, String password, Address address)
+ Victim()
+ Victim(UUID id, String firstName, String lastName, String username, String emailAddress, String password, Address address, ArrayList<Alert> recievedAlerts, byte age, char sex, String description, byte partySize, ArrayList<Request> createdRequests, ArrayList<Person> party, String phoneNumber)
+ Victim(User user)
+ getInfo(): String
+ getFullParty(): ArrayList<Person>
+ addPartyMember(Person person): void
+ getPerson(int index): String
+ notifyShelter(Shelter shelter, int partySize): void
+ createRequest(byte severity, Address address, String description): void
+ editRequestSeverity(Request request, byte severity): void
+ editRequestAddress(Request request, Address address): void
+ editRequestDescription(Request request, String description): void
+ toString(): String

Admin

- phoneNumber: String

+ Admin(String firstName, String lastName, String username, String emailAddress, String password, Address address, String phoneNumber)
+ Admin(User user, String phoneNumber)
+ Admin(UUID id, String firstName, String lastName, String username, String emailAddress, String password, Address address, ArrayList<Alert> recievedAlerts, String phoneNumber)
+ addHurricane(Hurricane hurricane): void
+ removeHurricane(Hurricane hurricane): void
+ updateHurricaneAddress(Hurricane hurricane, Address address): void
+ updateHurricanePath(Hurricane hurricane, ArrayList<Address> path): void
+ sendAlert(Alert alert): void
+ verifyCredential(Credential credential): void
+ toString(): String

Person

- name: String
- age: byte
- isPet: boolean
- description: String

+ Person(String name, byte age, boolean isPet, String description)
+ isPet(): boolean
+ toString(): String

Dispatcher

- phoneNumber: String

+ Dispatcher(String firstName, String lastName, String username, String emailAddress, String password, Address address, String phoneNumber)
+ Dispatcher(User user, String phoneNumber)
+ Dispatcher(UUID id, String firstName, String lastName, String username, String emailAddress, String password, Address address, ArrayList<Alert> recievedAlerts, String phoneNumber)
+ respondToRequest(Request request, Volunteer volunteer): void
+ requestMoreInfo(Request request, String whatIsNeeded): void
+ updateRequest(Request request, ResponseStatus status): void
+ finishRequest(Request request): void
+ toString(): String

ShelterWorker

- shelter: Shelter

+ ShelterWorker(UUID id, String firstName, String lastName, String username, String password, String emailAddress, Address address, ArrayList<Alert> recievedAlerts, Shelter shelter)
+ ShelterWorker(String firstName, String lastName, String username, String password, String emailAddress, Address address, Shelter shelter)
+ ShelterWorker(User user)
+ updateResources(Resource resource, int quantity): void
+ transferResources(Resource resource, Shelter shelter): void
+ finishRequest(Request request): void
+ toString(): String